

AI-Driven Attacks, Respective Cryptographic Risks and the Evolution of Cryptographic Defences

A Research report aims to analyse the development of cryptography as a method to tackle threats arising from attacks fueled by Artificial Intelligence.

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Section 1: Introduction

In the present modern digital world, where Artificial Intelligence has significantly evolved in various domains, the requirement of Cybersecurity has also increased. Cybersecurity can be defined as a practice that protects systems, networks, and electronic data from unauthorized access and malicious attacks.

Whereas cryptography is the science of security-based algorithms and safeguards information integrity, information source authenticity algorithms, and information confidentiality algorithms (encryption algorithms). It is the way to achieve security of network or data by hiding the important information from unauthorized individuals (Mustafa M. Abd Zaid, 2020) and hence enables secure storage and transmission of information in digital environments by employing cryptographic techniques and secure communication protocols.

Amid the growing complexity of cyber threats, especially those linked to artificial intelligence, advancing cryptography is crucial to combat these challenges. This report aims to analyse the development of cryptography as a method to tackle threats arising from attacks fueled by Artificial Intelligence.

While AI-driven attacks have increased the scale, speed, and sophistication of cyber threats, they do not directly break modern cryptographic algorithms such as AES, RSA, or ECC. Instead, AI primarily exploits weaknesses surrounding cryptographic systems — including identity compromise, social engineering, protocol misuse, deepfake-based authentication bypass, and automated vulnerability discovery. This means the core mathematical foundations of cryptography remain strong, but the cryptographic environment has become vulnerable. Therefore this study on the solutions of evolutionary requirements of modern cryptographic environment instead of revolutionizing it completely (complete reinvention of new algorithms) and discusses how cryptography is deployed, managed, and integrated (such as crypto-agility, AI-supported anomaly detection, and stronger authentication). Post-quantum cryptography is the only major algorithmic revolution, and it is driven by quantum computing rather than AI.

1.1 Background of Cybersecurity in the AI Era

Due to the growing reliance on internet technologies, cloud computing, and interconnected systems, cybersecurity has emerged as a crucial element in contemporary digital infrastructure. The contemporary world depends on digital platforms for communication, financial dealings, and data storage, leading to the challenge of safeguarding sensitive information against current cybersecurity threats. These cybersecurity threats have changed significantly with the rise of Artificial Intelligence.

The increasing integration of AI in cybersecurity has dramatically altered the cybersecurity landscape, making traditional techniques insufficient to handle these threats and protect digital systems. To safeguard digital systems against these emerging threats, advanced cryptographic methods, intelligent defensive strategies, and adaptive security systems are essential, known as the “AI Era” in cybersecurity.

1.2 Importance of Cryptography in Modern Digital Systems

According to the historical development of cryptography, we may classify cryptosystems to two groups based on their security objectives: conventional cryptography which concerns with privacy only; and modern cryptography deals with the insurance of confidentiality, integrity, authenticity and other objectives needed by variety of modern applications to ensure the protection of sensitive information. (Chan, 2005)

To ensure these fundamentals of cryptographic protocols some of the modern cryptographic techniques which are used for cybersecurity are SSL technology, TLS handshake, and end-to-end encryption, which are used to ensure data security from tempering, interception, and illegal stealing of data by hackers. Hash Functions and

Digital certificates are also used to ensure validation and verification of the identities of the communicating parties for increased security.

But due to the emergence of AI-based cyber attacks, it is observed that the challenges to traditional cryptographic systems have been evolved significantly, and the need to strengthen these cryptographic techniques to withstand these attacks and to develop advanced security mechanisms, it has increased.

1.3 Rising Role of Artificial Intelligence in Cyber Attacks

AI technologies were developed in the early stages with the aim of improving security systems. Nowadays, cybercriminals are using AI technologies to increase the level, rate, and sophistication of cyber-attacks and cybercriminals can automate complex attack processes in a targeted and adaptable manner.

Some important keyways by which AI enables more sophisticated cyberattacks are listed below:

- AI-powered malware and ransomware
- AI generated Social Engineering and Phishing Attacks
- Automated Vulnerability Discovery by machine learning algorithms
- Deepfake and identity spoofing
- Adversarial AI attacks

These attacks generated by the AI pose a threat to the Digital system and hence increase the need for incorporating more robust cryptographic principles in the cybersecurity field. The details of the attacks, the threat they pose, and the defensive strategies in the context of cryptography are discussed in the subsequent sections of this document.

1.4 Research Problem: AI-Driven Threats to Cryptographic Security

As discussed above in introduction section, the integration of AI has increased the sophistication, speed and scale of cyber attacks and hence building pressure for cryptographic ecosystem by exploiting the weaknesses (identity, key management, implementations, protocols).

AI-powered phishing, malware, deepfakes, automated vulnerability discovery, and adversarial attacks increase the likelihood of credential theft, protocol misuse, key exposure, and cryptographic bypass, creating indirect but significant threats to confidentiality, integrity, and authentication due to their power of adaptability, scalability and automation capability as AI can analyse large volumes of data, identify system vulnerabilities, and execute attacks with greater efficiency and precision. The research problem, therefore, is to examine how AI-driven attacks reshape the cryptographic security landscape not by breaking encryption directly, but by undermining the surrounding systems and to determine what evolutionary improvements in cryptographic practices are required to maintain confidentiality, integrity, and authentication in the AI era.

1.5 Objectives of the Study

The main objectives of this study are:

- To examine the growing role of artificial intelligence in modern cyber attacks by changing cryptographic landscape through identity compromise, protocol misuse and implementation level vulnerabilities.

- To evaluate the evolution of cryptographic defence strategies developed to counter AI-based cyber threats, such as: strong authentication, crypto-agility, anomaly detection, and secure key management
- To highlight future approaches and technologies(PQC and AI based defensive mechanism) aimed at strengthening cryptographic security in the AI era.

1.6 Structure of the Report

This report is organized into seven sections:

The first section highlight introduction along with research objectives and research problem to clarify the purpose and direction of this report.

The second section of this report focuses on the key definitions and foundational conceptual terminologies used in this research to help the readers understand subsequent analysis more effectively.

Third section of this report deals with the analysis of cryptographic risks associated with each kind of AI Driven Attack and outlines respective defence mechanisms separately for each attack category in this research.

Fourth section of this report discusses how cryptographic practices are adapting to AI-driven threats through stronger authentication, dynamic cryptographic systems, and AI-assisted protective measures.

Fifth section of this study deals with the case studies to illustrate some real-world scenarios of AI Driven attacks and how they defended in practical environments.

Sixth section of this report highlights the future approaches designed to address AI driven attacks and the conclusion of this report.

Seventh Section of this report includes the references of this study.

Section 2: Conceptual Foundation and Key Terms

2.1 AI-Driven Attacks

AI Driven Attack refers to the cyberattacks that are created, enhanced or automated by Artificial Intelligence. These attacks are more sophisticated, and difficult to defend. These attacks may use AI techniques such as analysis of large volume of data , pattern recognition, natural language generation, decision-making algorithms, and adaptive learning, to improve the effectiveness, precision, and scalability of malicious activities.

These attacks may include adaptive malwares, deepfake identity spoofing, adversarial AI Attacks, AI generated social engineering attacks, automated vulnerability discovery by targeting cybersecurity. These attacks are not direct threat to cryptographic algorithms instead these attacks exploit vulnerabilities of system (identity verification, protocol implementation, key management, and human-centric processes) and hence increasing the risk of cryptographic by-pass or unauthorized access.

2.2 Cryptanalysis

Cryptanalysis is the technique of identifying vulnerabilities in cryptographic algorithms by identifying flaws and weaknesses in them. These vulnerabilities enable the breaking of these algorithms and reveal secret information such as keys or cipher text. In modern cybersecurity it also includes analysis for the exploitation of implementation weaknesses (flaws led by incorrect execution of algorithm in hardware/software) or side channel leakages (sensitive information leaks through unintended physical signals such as timing, power consumption, or electromagnetic emissions during cryptographic operations without breaking encryption algorithm itself).

2.3 Adaptive Malware

Adaptive malware or simply AI Driven malware refers to the kind of malicious software that uses AI or machine learning techniques to change its behavior, pattern and appearance to bypass the security detection. These malwares actively learn from the environment and changes themselves to legitimate software whenever they are likely to be detect and hence avoid signature-based or behavioral-based detection.

2.4 Zero-Trust Architecture

Zero-Trust Architecture is a security model based on the principle “never trust, always verify,” meaning no user, device, or application is trusted by default, even if already inside the network. Every access request must be continuously authenticated, authorized, and validated using strong identity verification, least-privilege access, and ongoing monitoring to reduce the risk of breaches and cryptographic misuse.

2.5 Post-Quantum Cryptography

Post Quantum cryptography is a new class of cryptography that deals with the encryption algorithms that use mathematically hard problems like lattice based, hash based or code-based construction.

PQC ensures that when future quantum computers gain the ability to break current encryption standards (e.g., RSA, Elliptic Curve), sensitive data will remain secure. PQC is therefore essential for shielding data from malicious parties, for complying with future data regulations, and for safeguarding online data privacy and protections like TLS. (Cloudflare, n.d.)

2.6 Crypto-Agility:

Crypto agility is system’s capability that allows it to interchange or upgrade cryptographic algorithms without requiring major changes in architecture. It enables rapid replacement of weakened or outdated cryptographic schemes in the presence of new vulnerabilities detection such as AI driven attacks. For example, migrating from RSA to PQC Kyber demonstrates crypto agility in practice.

Section 3: AI-Driven Attacks and Their Cryptographic Security Risks

Some main characteristics of AI driven attacks are that they are faster, scalable, automated, adaptive, personalized and more précised as compared to conventional attacks and hence involve new cryptographic risks.

This section presents the analysis of the AI driven attacks outlined in introduction section focusing on their behavior(definition), cryptographic risks and required cryptographic defenses to mitigate them.

3.1 Type 1: AI- Generated Phishing & Social Engineering

AI generated phishing and social Engineering attacks use artificial intelligence by exploiting its capability of Large Language Model and Generative AI to design highly convincing, stylistically accurate and context aware messages and emails that appear to be legitimate. These messages exploit human trust or create a sense of urgency which eventually helps the attacker in authentication bypass and cryptographic protections failure without breaking underlying algorithms and might result in credential theft, identity compromise, and unauthorized access.

3.1.1 Cryptographic Risks

3.1.1.1 Compromise of Authentication & Digital Identity Systems

These attacks trick users to reveal their OTPs, digital certificates, private keys and login credentials by designing personalize, human deceptive messages or emails which result in credentials compromise eventually allowing the attacker to enter in the system as legitimate user and bypass cryptographic access control.

3.1.1.2 Interception of Encrypted Communications via Social Engineering

By creating highly convincing impersonation or authority-based deception attacks generated by AI, attackers can trick the users for sharing encrypted documents exchanging sensitive information over insecure channels, approving malicious certificate requests, or disabling security features. Which results in indirect data exposure instead of cracking algorithmic mechanism.

3.1.2 Cryptographic Defenses

3.1.2.1 Strong Cryptographic Authentication (MFA, Digital Signatures)

Use of strong multifactor authentication mechanisms, hardware backed keys FIDO2 and digitally signed authentication requests mechanisms help in securing the system from such risk factors. This stops attackers entering the system without authentication of possession-based elements such as: hardware security key, sim-based OTPs, soft tokens. As a result, even if users fall victim to AI-generated phishing attempts, the attacker cannot complete the authentication process without the cryptographic proof of possession.

3.1.2.2 Secure Email Encryption & Certificate-Based Validation

End-to-end email encryption, certificate-based identity validation, and DKIM (domain key identified mail)/SPF (sender policy framework)/DMARC (Domain-based Message Authentication, Reporting and Conformance) verification help prevent spoofed emails from appearing legitimate, reducing the success rate of AI-generated phishing attempts.

3.1.2.3 AI-Based Detection in Encrypted Environments

Integration of AI-assisted email filtering tools and AI-based anomaly detection models which helps the system to analyze the pattern of communication and email characteristics to identify suspicious or abnormal behavior which reduces the chances of users being victimized of such attacks.

3.2 Type 2: AI- Enhanced Malware & Ransomware

AI enhanced malware and ransomware are especial type of malicious software that actively learn from their existing environment and adapt their behavior. These malwares can modify their code, mimics legitimate

behavior and alter their signature in real time to evade detection mechanism. This capability help them to maintain longer persistence and make their detection difficult for traditional tools.

3.2.1 Cryptographic Risks

3.2.1.1 Abuse of Encryption by AI-Driven Ransomware

AI Driven ransomwares are special ransoms that uses artificial intelligence to detect sensitive data, speeding up the key generation and distribution and selecting optimal encryption routines. Which locks the files and databases more efficiently with high speed and greater precision, making data recovery very difficult for the system without decryption key.

3.2.1.2 Exploitation of Weak Cryptographic Implementations

AI Driven malwares can analyze systems for weak cryptographic implementations such as poor randomness, improper key storage, reused nonces, deprecated cipher suites. These weaknesses allow the malware to by-pass cryptographic security by simply exploiting implementation flaws rather than breaking algorithm themselves.

3.2.1.3 Evasion of AI-based Security & Encrypted Network Monitoring

AI based malware dynamically change their network behavior (like time interval for communication, packet size and traffic patterns etc.) which disguise the malicious encrypted message (send to attacker's command and control system) as a legitimate traffic and hence making its detection more challenging for traditional network security systems and encrypted traffic monitoring tools.

3.2.2 Cryptographic Defenses

3.2.2.1 Secure Cryptographic Key Management

The use of secure key management practices can prevent ransomware from encryption, decryption or misuse of sensitive data. These practices include hardware security modules (HSMs), secure key storage, frequent key rotation, and strict access control which will prevent the ransomware from obtaining or manipulating encryption keys.

3.2.2.2 Layered Encryption & AI-Enhanced Malware Detection

Using multiple layers of encryption to encrypt data at different stages (data at rest, data at transient) along with AI enhanced threat detection tools improves security against AI driven malwares. These AI enhanced malware detection tools continuously monitors system behavior and helps in early detection of ransomware by checking unusual encryption activity, rapid file-modification patterns, or abnormal key-access attempts.

3.2.2.3 Zero-Trust Architecture for Encrypted Systems

Zero Trust Architecture is a security framework build upon the framework of "never trust always verify" (Fortinet, n.d.). It prevents the lateral movement of malware within the system which stops the malware to reach sensitive data and cryptographic systems due to network segmentation, Continuous verification, least-privilege access and strict authentication controls.

3.3 Type 3:Automated Vulnerability Discovery (AI-Based)

Automated vulnerability discovery refers to the integration of AI in vulnerability discovery process which makes it easier to find cryptographic implementation flaws, coding errors and misconfigurations within

software or systems. Once these weaknesses are found, attackers can exploit them to gain unauthorized access, extract keys, manipulate encrypted communications, or bypass cryptographic protections

3.3.1 Cryptographic Risks

3.3.1.1 3.3.1 AI-Assisted Cryptanalysis

AI assisted cryptanalysis help the attackers to automate the process of finding implantation flaws, weak configurations and side channel vulnerabilities (e.g. timing and power-trace leakages). AI reduces human efforts by analyzing the pattern discovery and anomaly detection at faster speed and large scale, which help in algorithm exploitation instead of breaking it, even if the algorithm is mathematically strong.

3.3.1.2 Exposure of Weak Cryptographic Implementations

AI based Machine learning can expose weak cryptographic algorithms automating security analysis at very fast speed by analyzing the cryptographic implementation mistakes. ML can scan source code(codebases) and compiled codes(binaries) for security mistakes by checking cryptographic implementation flaws like: poor randomness, nonce reuse ,deprecated cipher suits and weak key storage and finally make a prioritize list i-e triage list with potential security issues mentioned eliminating the need of manual security audits.

3.3.1.3 Harvest-Now-Decrypt-Later Risks

Some secrets remain valuable for many years, even if an adversary can't crack the encryption that protects our secrets at the moment, it could still be beneficial to capture encrypted data and hold onto it, in the hopes that a quantum computer will break the encryption down the road. This idea is sometimes expressed as "harvest now, decrypt later" (NIST, 2025)

Automated vulnerability discovery allow attackers to identify valuable data repositories in encrypted form and system weaknesses on a large scale. Which rise to the issue of "Harvest now decrypt later" strategy raising long-term confidentiality risk for sensitive archives and regulated data

3.3.1.4 Threats to Confidentiality & Data Integrity

By rapidly identifying implementation flaws in the code and misconfigurations in the protocol, AI-driven discovery helps in increasing the chances of unauthorized decryption, key compromise, or even forgery paths, even if the underlying algorithms are mathematically sound.

3.3.2 Cryptographic Defenses

3.3.2.1 Adaptive Cryptographic Algorithms & Crypto-Agility

The concept of crypto agility provides rapid remedy against cryptographic weaknesses discovered by AI in any existing cryptographic configuration. It allows the system to adapt and upgrade algorithms, key size and parameters without redesigning the whole system.

3.3.2.2 Post-Quantum Cryptography for Long-Term Protection

To handle HNDL risk and future quantum attacks, encryption methods that can stave off cyberattacks by both the conventional computers of today and the quantum computers of tomorrow is needed. NIST is leading a global effort to create electronic defences against such attacks through its Post-Quantum Cryptography (PQC) project, which released the first three finalized PQC standards in 2024 (NIST, 2025).

3.3.2.3 Continuous Security Auditing & Cryptographic Hardening

Along with secure code review and red team testing, the integration of AI assisted Static Application Security testing and Dynamic Application Security Testing to CI/CD pipeline for continuous and automatic scanning of code reduces exploitable attack surface by faster detection of cryptographic misuse.

3.4 Type 4: Deepfake & Identity Spoofing Attacks

Deepfake and identity spoofing attacks use generative AI for create highly convincing fake audio, videos and images that appear legitimate. Such attacks do not break cryptographic algorithms directly, they undermine the identity layer that cryptographic systems rely on by exploit human trust and impersonating individual's behavior.

3.4.1 Cryptographic Risks

3.4.1.1 Compromise of Biometric Authentication Systems

Deepfake identity spoofing attacks are a major threat to biometric verification as they can manipulate the identity verification to exploit weak or poorly configured biometric checks without robust liveness or injection-attack detection which enables the attacker for unauthorized logins and session takeovers.

3.4.1.2 Identity Fraud in Secure Digital Platforms

In secure digital platforms, cryptographic access control can be bypassed through deepfake impersonation attacks where the attacker can manipulate remote verification system. As a result, the attacker can gain access to system as a legitimate user with valid token or keys.

3.4.1.3 Manipulation of Trust in Digital Transactions

Deep fake identity spoofing can bypass cryptographic security even if the messages are encrypted and digitally signed, as it manipulates human decision-making and convince victim to authorize fraudulent transactions with the help of synthetic(fake AI generated) audio/video message.

3.4.2 Cryptographic Defenses

3.4.2.1 Cryptographic Identity Verification & Digital Certificates

In systems, identity of user is verified by digital certificate such as those used in mutual TLS (mTLS), client certificate authentication, or S/MIME where appropriate. When this verification is bound to specific registered device, the requests of digital certificates are accepted by only trusted device (ensuring chain of trust).

3.4.2.2 Multi-Factor + Biometric- Backed Cryptographic Authentication

Another defense strategy against such attacks includes multifactor authentication using hardware backed keys and liveness checked biometrics. Due to this approach and only fake biometric or stolen passwords are not sufficient for account logins and by-pass security instead, it needs approval through trusted device.

3.4.2.3 Blockchain-Based Identity Validation & Signature Verification

Blockchain-based identity and signature verification is a security approach in which public keys, credential status, and revocation records are stored on an immutable ledger and for the verification of transaction firstly identity confirmation and key validation takes place through ledger and then digital signature is verified using the corresponding public key which helps detect impersonation attempts and provides tamper-evident auditability of signatures and identity updates.

3.5 Type 5:Adversarial AI Attacks

An adversarial AI attack is a malicious technique that manipulates machine learning models by deliberately feeding them deceptive data to cause incorrect or unintended behavior. These attacks exploit vulnerabilities in the model's underlying logic, often through subtle, imperceptible changes to the input data. They challenge the trustworthiness and reliability of AI systems, which can have serious consequences in applications such as fraud detection, autonomous vehicles, and cybersecurity. (PaloAltoNetworks, n.d.)

3.5.1 Cryptographic Risks

3.5.1.1 Manipulation of AI-Based Security Systems

Adversarial AI attacks that lie under the category of Evasion attacks are a big threat to ML models. These attacks can craft input to AI Based security tools (like: anomaly detection, malware classifiers, or encrypted-traffic monitoring systems) and manipulates it to misclassify malicious activity as benign causing wrong cryptographic privileges.

3.5.1.2 Bypassing Intrusion Detection & Encrypted Monitoring

Adversarial AI Attacks can exploit the weaknesses in ML Model and encrypt the malicious traffic as normal encrypted traffic through changing in packet size, time and pattern. Which helps in by passing the intrusion detection security,enabling covert channel creation, encrypted data exfiltration, or unauthorized command- and- control communication.

3.5.1.3 Data Poisoning of Training Models

Data poisoning of training models allows attackers to inject malicious data into training datasets. This corrupts the model's learning process, leading to degraded accuracy or deliberate vulnerabilities. (SysDig, n.d.). This does not directly affects cryptography instead, it weakens the AI based security monitoring use for cryptographic protection.

3.5.2 Cryptographic Defenses

3.5.2.1 Adversarially Robust AI Security Models

To make AI supported cryptographic protection more effective, reliable and resilient to these adversarial attacks, some techniques that might help include adversarial training (where model is deliberately trained against deliberately crafted deceptive example), gradient masking(making it difficult for the attacker to understand model working),model hardening and continuous retaining should be considered.

3.5.2.2 Cryptographic Integrity Protection for Training Data

Another mitigation of such attacks is that before training the AI model, the integrity and originality of training data should be verified via digital signatures, hash-based verification or secure provenance tracking (tracking data history). This cryptographic integrity eliminates the poisoned data insertion chances and improves the reliability of cryptographic enforcement mechanisms that depend on AI-supported monitoring

3.5.2.3 Secure & Explainable AI for Cyber Defense

To reduce the risk of adversarial manipulation Explainable AI techniques are used due to which AI decisions are made explainable for human review, securely recorded in tamper-resistant logs, and restricted by strict access controls. As a result, no single AI output can directly grant sensitive cryptographic privileges. This help in reducing the poisonous model training and false decision making.

Section 4: Evolution of Cryptographic Defenses in the AI Era

While section 3 of this report have already discussed the cryptographic risks and defenses against each kind of AI Driven Attack, this section will only highlight 5the broader illustration of cryptographic evolutions in AI Era by outlining the overall shift without repeating the explanation that have already been discussed above. This evolutionary shift can be concluded as:

- The focus of cryptography is shifting from purely algorithmic strength to **stronger identity-based controls** such as MFA, FIDO2, and digital signatures due to AI-enhanced phishing.
- **Improved key-management practices**, securing private keys, randomness, and implementation hygiene is an important need of cryptographic security now a days to resist AI-assisted malware and exploitation.
- **Up-to-date protocols** (e.g., TLS 1.3, secure cipher suites) to counter AI-accelerated vulnerability discovery in outdated or weak cryptographic implementations is also a requirement of Modern System in this AI Era.
- **AI-assisted anomaly detection** is becoming essential to monitor encrypted traffic patterns and identify evasion attempts that traditional tools fail to detect.
- Zero-trust principles have become essential due to AI-enabled bypass methods.
- Crypto-agility is now necessary because AI accelerates vulnerability discovery.

Section 5: Case Studies: AI vs Cryptography in Practice

The relevance of these findings is supported by numerous real-world incidents involving AI-generated phishing, adaptive ransomware, deepfake-based identity fraud, and AI-automated vulnerability exploitation. Some references of case studies are also attached in this section for reference to validate the facts.

MITRE ATLAS (Adversarial Threat Landscape for Artificial-Intelligence Systems)⁷ provides a structured framework for understanding how AI technologies can be exploited by adversaries, including for autonomous malware generation. . By formalizing these tactics within a matrix similar to MITRE ATT&CK, ATLAS contributes a valuable perspective on emerging AI-enabled threats and supports the development of targeted defensive strategies in response to AI-generated malware. (Zdrojewski, 25 September 2025).

- **Reuters Investigation (2025):** Demonstrated how major AI chatbots helped craft persuasive phishing emails used to scam elderly volunteers.
- **CyberHawk/PhishGPT Case (2025):** Showcased multi-channel AI-generated phishing including deepfake Zoom calls and voice cloning in real attacks (Consultancy, 2025)
- **AI-Powered Ransomware “PromptLock” (2025):** First confirmed AI ransomware generating dynamic scripts using LLMs, discovered by ESET researchers (News, 2025)
- **\$25M Arup Deepfake Fraud (2024):** Attackers used AI-generated video and voice deepfakes of the CFO and team in a live call, leading to \$25.6M loss. (VerifyReal, 2024)

- **Chinese GTG- 1002 AI- Orchestrated Cyber Espionage Campaign (2025):** Attackers used Anthropic’s Claude to autonomously conduct reconnaissance, exploitation, lateral movement, and exfiltration. (Disrupting the first reported AI-orchestrated Cyber Espionage Campaign, November 2025)
- **HexStrike- AI Being Used to Exploit Citrix Zero- Days (2025):** Attackers weaponized an AI pentesting tool to exploit n-day flaws within hours (Computer, 2025)
- **AI-Powered Cyberattacks: A Comprehensive Review and Analysis of Emerging Threats:** This research paper shows a comprehensive analysis over most of the cyber attacks covered in this report, to show the impact on Modern Cybersecurity. (Zdrojewski, 25 September 2025)

Section 6: Conclusion

This report suggests that AI-driven attacks usually do not directly break cryptographic algorithms. Instead, they target the surrounding cryptographic ecosystem, such as identity systems, key management, protocol configurations, and implementations. By using scalable, adaptive, and automated techniques—like AI-generated phishing, AI-enhanced malware or ransomware, automated vulnerability discovery, deepfake-based spoofing, and adversarial machine learning—attackers are able to exploit weaknesses around cryptography rather than the algorithms themselves. Consequently, the response does not need to be entirely revolutionary but rather evolutionary. This may requires:

- Organizations should plan and gradually implement post-quantum cryptography (PQC), especially for long-term data and critical communications. This includes using hybrid approaches, keeping track of algorithms, keys, and certificates, testing performance, and rolling out changes in phases.
- Identity must be tightly linked with cryptographic security by using phishing-resistant multi-factor authentication (such as FIDO2) and, where needed, biometric liveness checks. This ensures that approvals are tied to trusted devices and protected keys, so deepfake attacks alone cannot authorize actions.
- It is important to adopt crypto-agility, allowing systems to quickly update or replace algorithms, keys, and parameters across applications and services whenever new vulnerabilities or standards emerge.
- AI should also be used defensively by deploying intelligent monitoring systems that can detect anomalies in encrypted traffic, track key usage, and maintain certificate security. For sensitive actions, human oversight should be included alongside AI decisions.
- The machine learning pipeline that supports cryptographic security must be protected by verifying datasets and models, tracking their origin, and validating updates to prevent poisoning attacks. Techniques like adversarial training and explainability should be used so that no single AI decision can grant high-level cryptographic access without verification.
- Finally, zero-trust principles should be consistently applied, including continuous verification, least-privilege access, network segmentation, and device attestation, to limit the impact if identities or systems are compromised through AI-driven social engineering.

*Together, these steps ensure that cryptography remains dependable in an AI-rich threat environment—by evolving how it is **applied, integrated, and governed**, not by reinventing its mathematical foundations.*

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